Bayesian modelling of the tree height-diameter relationship

Project report

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Contents

1 Problem Statement

The aim of this project is to explore using Bayesian methods the relationship between the height of a tree of a particular species and its diameter measured at breast height. This problem is crucial in forestry for multiple reasons.

As will be shown, height of a tree is a very important metric, though its measurement may cause practical difficulties (for higher trees foresters are required to use specialized devices, which still do not always guarantee perfect precision, apply less accurate estimation methods or climb directly onto the tree), while the measurement of the diameter is straightforward. The predictive properties of the proposed model might help further estimation of the height based on precise diameter measurements.

Height of a tree, as useful as it can be on its own, is also essential in calculating derivative parameters such as log volume (volume of wood in an individual tree) or stand volume (in the case of a group of trees) which, in turn, make it possible to determine the forest's resource potential. The other applications include the calculation of biomass of a tree and consequently evaluating the storage of carbon dioxide in the forest, as well as soil quality (bonitation) assessment.

For these reasons, the subject of this project is already quite well established, with the height-diameter curve as one of the most important and practical tools in forestry. However, the readily available studies involve problems and solutions that slightly differ from the ones presented in this report (examples would be [2] - comparison of multiple models in frequentist setting, [3] - comparison of Bayesian non-hierarchical models, [4] - mixed-effects (hierarchical) Bayesian models with breast-height age as explanatory variable, different link function).

Finally, red spruce (*Picea rubens*), the species selected for modelling, is known for its remarkable wood, used for acoustic stringed instruments, and in the paper pulp industry. It is also a popular choice for a holiday tree and usual lumber.

2 Data Description

The data used in this project comes from the official website of the Forest Service, U.S. Department of Agriculture, or more precisely, the Forest Inventory and Analysis program (FIA) of the USDA Forest Service Research and Development Branch, best accessible on the website [7]. We selected the state of Vermont as the area of interest and then downloaded a csv file VT_TREE.csv containing the necessary data. The state selection was strictly related to the choice of species, which in turn was based on the similarity between the red spruce, found in North America, and the Norway spruce (*Picea abies*), found in Europe and the only species of spruce found in Poland.

Given the large number of variables in the data frame, we had to choose only the ones relevant in the context of the research question and proposed model. To that end, we used The Forest Inventory and Analysis Database User Guide (provided by FIA along with the data), which helped us detect the essential variables and understand the structure of the data (extract of this guide with detailed explanation of the selected variables is attached to this report, file data_info.pdf). The key steps in data processing included following specifications:

- species selection
- diameter measured at breast hight, rather than at the root collar
- only live trees (this choice is quite arbitrary, though omitting dead trees simplifies the interpretation; potential issues closely related to the measurements of dead trees, such like measuring broken trees, are addressed in the following specifications)
- only unbroken trees (that is, the actual height is equal to the total height, where total height in case of a broken tree is an estimation of its height as if it were unbroken)
- accurate field measurements of both total and actual heights, rather than visual assessments or estimates based on a model (for the sake of modelling precise measurements should be preferred,

since estimated records can introduce additional noise and unceratainty; because accurate data is abundant, we omit the estimated values); in practice, this is a stronger requirement than the previous specification

- 2022 as the main inventory year considered in our project; years 2012 and 2017 for reference
- making sure that no tree was measured twice, by grouping the trees by survey unit code (survey units are usually groups of counties within each state), county code and plot number, and identifying each tree by the tree number; this shows the hierarchical structure of the data, in particular division of the state into counties, and of each county into plots of land.

The next step involved transformations of the quantitative variables. The height, which was originally given in feet, has been converted to meters and the diameter at breast height, given in inches, has been converted to centimeters. The final preparation for modelling consisted of applying the natural logarithm to both of these variables, calculating summary statistics and plotting scatter plots before and after log-transformation.

	inventory_year	survey_unit	county	plot_number	tree_number	diameter	height	SPCD	log_height	log_diameter
count	344.0	344.000000	344.000000	344.000000	344.000000	344.000000	344.000000	344.0	344.000000	344.000000
mean	2022.0	2.546512	13.854651	1082.069767	17.476744	19.069198	12.493256	97.0	2.372651	2.745171
std	0.0	0.498557	9.336085	929.084954	15.193006	10.793051	6.040294	0.0	0.605479	0.704223
min	2022.0	2.000000	1.000000	17.000000	1.000000	2.540000	2.133600	97.0	0.757811	0.932164
25%	2022.0	2.000000	5.000000	333.000000	6.000000	12.890500	7.620000	97.0	2.030776	2.556454
50%	2022.0	3.000000	9.000000	852.000000	13.000000	17.526000	12.192000	97.0	2.500780	2.863685
75%	2022.0	3.000000	23.000000	1468.000000	23.000000	25.146000	16.764000	97.0	2.819234	3.224699
max	2022.0	3.000000	27.000000	3337.000000	73.000000	62.230000	26.517600	97.0	3.277809	4.130837

Figure 1: Summary statistics.

Judging by Figure 1, nothing seems amiss. The shortest tree is just above 2 meters (which is reasonable, since its diameter had to be measured at breast height, which is about 1.3m), while the highest is 26.5 meters tall. The mean and the median of the height are close to each other, at around 12.5m and 12.2m respectively. The slimmest tree has a diameter of 2.54cm, or equivalently 1 inch, which would suggest that it was most likely rounded up (but the value must have been positive). The diameter of the thickest tree is 62.23cm, while the mean and the median are around 19.1cm and 17.5cm respectively.

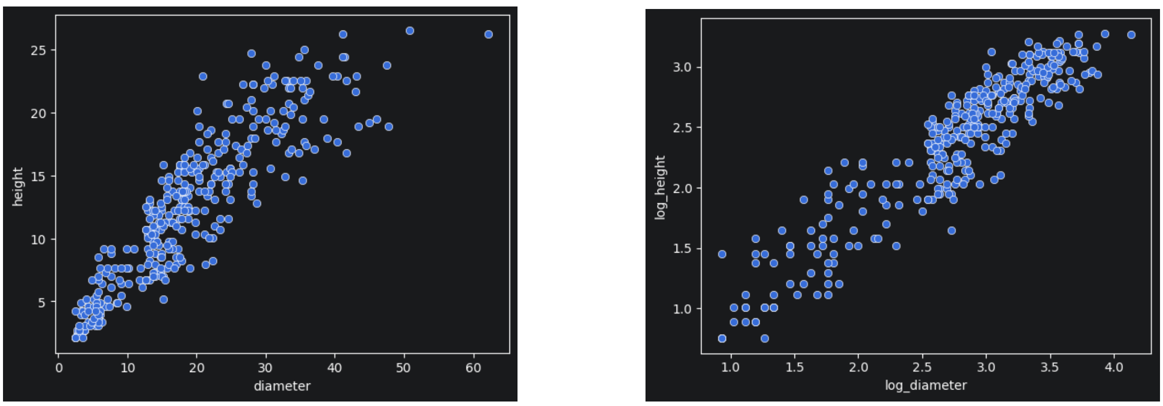

Figure 2: Scatter plots before and after log-transformation.

Figure 2 provides a visual representation of the relationship between these two variables. The plot on the left might suggest heteroscedasticity of the data (that is, the larger the diameter, the greater the variance in height measurements), while this property is not visible on the second plot. In addition, the plot on the right indicates a linear relationship between the log-transformed variables, which further prompts us to include this transformation in our model.

3 Model Description

3.1 General formulation of the model

We assume that the relationship between height and diameter is given by a power function, that is

$$\text{Height} = e^{\alpha} \cdot \text{Diameter}^{\beta} \cdot e^{\varepsilon},$$

where $\varepsilon \sim N(0, \sigma)$ is independent from Diameter and α, β and σ are unkonown parameters we want to estimate. After transforming both sides by logarithm this yields us linear relationship

$$\text{LogHeight} = \alpha + \beta \cdot \text{LogDiameter} + \varepsilon.$$

At this point we can provide interpretation for the coefficients α, β and anticipate their values. Namely, α should be approximately equal to the natural logarithm of the height, when the diameter is one unit, though it should be noted that we cannot simply transform it back and interpret it as such, since generally the expectation of logarithm is not equal to the logarithm of expectation, and some correction is required. However, this does not apply to β, which is the elasticity coefficient, meaning that a 1% change in diameter is associated with a β% change in height. Since naturally an increase in diameter corresponds to an increase in height this coeffiecient should be positive ($\beta > 0$). Based on previous studies, such as [8], where in one of proposed models this coefficient is even fixed ($\beta = 0.67$), we can expect that $\beta \in (0.4, 0.8)$ approximately. This means that the power function describing the relationship under study is concave ($0 < \beta < 1$), which is consistent with the scatter plot shown in Figure 2.

We consider three approaches that differ in the extent to which they utilize the hierarchical data structure, and in one case, based on the previous discussion, we additionally use an informative prior for the elasticity coefficient β.

3.2 Simple non-hierarchical model

In this setting we omit the hierarchical structure of our data and fit a Bayesian linear model of the form

$$\text{LogHeight}_i = \alpha + \beta \cdot \text{LogDiameter}_i + \varepsilon_i,$$

where $i = 1...n$ indexes trees and $\varepsilon_i \sim N(0, \sigma)$ are independent. The likelihood is of the form

$$\text{LogHeight}|\text{LogDiameter}, \alpha, \beta, \sigma \sim N_n(\alpha + \beta \cdot \text{LogDiameter}, \sigma^2 I).$$

Since we fit the model with the help of the Bambi Python package (BAyesian Model-Building Interface), we use default weakly informative priors (s.t. $p(\alpha, \beta, \sigma) = p(\alpha)p(\beta)p(\sigma)$, i.e. α, β and σ are independent a priori):

$$\alpha \sim N(2.3727, 6.0911) \qquad \beta \sim N(0, 2.1495)$$

$$\sigma \sim \textit{Half-Student-t}(4, 0.6046).$$

The density h of $\textit{Half-Student-t}(\nu, \sigma)$ is given by

$$h = 2f \cdot \mathbb{1}_{(0,+\infty)},$$

where f is the density of Student's t distribution with parameters ν (degrees of freedom), σ (scale) and location parameter $\mu = 0$.

Looking at the prior distributions for α and β, we can conclude that they are indeed weakly informative; in particular, the prior distribution for the elasticity coefficient β allows, with a fairly high probability, for values that are negative or greater than 1.

3.3 One-level hierarchical model with varying slope

We group our data by county and for each county $k = 1...m$ find county-specific intercepts α_k and slopes β_k. The model takes form

$$\text{LogHeight}_{i_k} = \alpha_k + \beta_k \cdot \text{LogDiameter}_{i_k} + \varepsilon_{i_k},$$

where $i_k = 1...m_k$ indexes trees in k-th county and $\varepsilon_{i_k} \sim N(0, \sigma)$ are independent.

In this setting we use Bambi-default weakly informative hyperpriors for the mean and covariance matrix of the joint prior multivariate normal distribution of coefficients α and β. Since it turns out that this model has weaker predictive capabilities, a detailed decomposition of the hierarchical structure will be presented in the next model formulation.

3.4 Two-level hierarchical model with non-varying slope

We group the trees by county, indexed by $k = 1...m$, and in each county by plot, indexed by $j_k = 1...m_k$. The hierarchical structure is taken into account only by the plot-specific intercepts α_{j_k}, the slope β is fitted for the whole state. The model is

$$\text{LogHeight}_{i_{j_k}} = \alpha_{j_k} + \beta \cdot \text{LogDiameter}_{i_{j_k}} + \varepsilon_{i_{j_k}},$$

where $i_{j_k} = 1...m_{j_k}$ indexes trees in j_k-th plot in k-th county and $\varepsilon_{i_{j_k}} \sim N(0, \sigma)$ are independent.

Observe that plot-specific intercepts can be decomposed:

$$\alpha_{j_k} = \alpha + \tilde{\alpha}_k + \tilde{\alpha}_{j_k},$$

where $\tilde{\alpha}_k$ and $\tilde{\alpha}_{j_k}$ describe group-level effects for counties and plots, respectively, and α is the global intercept.

As mentioned earlier, we consider two approaches:

- using weakly informative (default) priors for all parameters
- as before, but imposing an informative prior on β.

Since we already discussed the weakly informative prior for β in the case of the simple model, here we will focus on the model with the custom prior for β.

The prior specification for intercept reads:

$$\alpha \sim N(\mu_\alpha, \sigma_\alpha) \qquad \tilde{\alpha}_k \sim N(0, \tilde{\sigma}_{\alpha_k}) \qquad \tilde{\alpha}_{j_k} \sim N(0, \tilde{\sigma}_{\alpha_{j_k}})$$

$$\tilde{\sigma}_{\alpha_k}, \tilde{\sigma}_{\alpha_{j_k}} \sim \mathit{Half\text{-}Normal}(\sigma_\alpha)$$

where $\mu_\alpha = 2.3727,\ \sigma_\alpha = 1.5115$ are the fixed hyperparameters, provided by Bambi.

In other words

$$\alpha \sim N(\mu_\alpha, \sigma_\alpha) \qquad \tilde{\sigma}_{\alpha_k}, \tilde{\sigma}_{\alpha_{j_k}} \sim \mathit{Half\text{-}Normal}(\sigma_\alpha)$$

$$\alpha_k | \alpha, \tilde{\sigma}_{\alpha_k} \sim N(\alpha, \sqrt{\sigma_\alpha^2 + \tilde{\sigma}_{\alpha_k}^2})$$

$$\alpha_{j_k} | \alpha, \alpha_k, \tilde{\sigma}_{\alpha_k}, \tilde{\sigma}_{\alpha_{j_k}} \sim N(\alpha_k, \sqrt{\sigma_\alpha^2 + \tilde{\sigma}_{\alpha_k}^2 + \tilde{\sigma}_{\alpha_{j_k}}^2})$$

where $\tilde{\sigma}_{\alpha_k}$ and $\tilde{\sigma}_{\alpha_{j_k}}$ depict county-specific and plot-specific standard deviations, respectively, and $\alpha, \tilde{\alpha}_k, \tilde{\alpha}_{j_k}$ are independent.

For β (growth elasticity coefficient) we choose prior $\beta \sim N(0.6, 0.07)$ based on already established height-diameter curves. Prior for σ reads:

$$\sigma \sim \mathit{Half\text{-}Student\text{-}t}(4, 0.6046).$$

```
     Formula: log_height ~ log_diameter + (1 | county) + (1 | county:plot_number)
      Family: gaussian
        Link: mu = identity
Observations: 344
      Priors:
  target = mu
      Common-level effects
          Intercept ~ Normal(mu: 2.3727, sigma: 1.5115)
          log_diameter ~ Normal(mu: 0.6, sigma: 0.07)

      Group-level effects
          1|county ~ Normal(mu: 0.0, sigma: HalfNormal(sigma: 1.5115))
          1|county:plot_number ~ Normal(mu: 0.0, sigma: HalfNormal(sigma: 1.5115))

      Auxiliary parameters
          sigma ~ HalfStudentT(nu: 4.0, sigma: 0.6046)
```

Figure 3: Prior summary in Python.

3.5 Posterior approximation

In each setting, due to the complex specification of the priors, an exact closed-form formula for posterior distribution cannot be derived analytically (we know the corresponding density up to a constant), so we resort to sampling from the posterior using Markov Chain Monte Carlo methods. Bambi, similarly to the package rstanarm in R, uses the NUTS (No U-Turn Sampler) algorithm (which is an extension of the Hamiltonian Monte Carlo method) to sample from the posterior distribution. To reduce the autocorrelation in the sample we applied a thinning procedure.

4 Model Diagnostics

In this section we will discuss the use of basic diagnostic tools for Bayesian modelling in our project, without a detailed breakdown into different model formulations, since we obtained very similar results in every case.

4.1 MCMC convergence monitoring and autocorrelation assessment

To determine whether convergence had been reached we used, wherever it was practical, trace plots and, most commonly, $\hat{R}$ statistic.

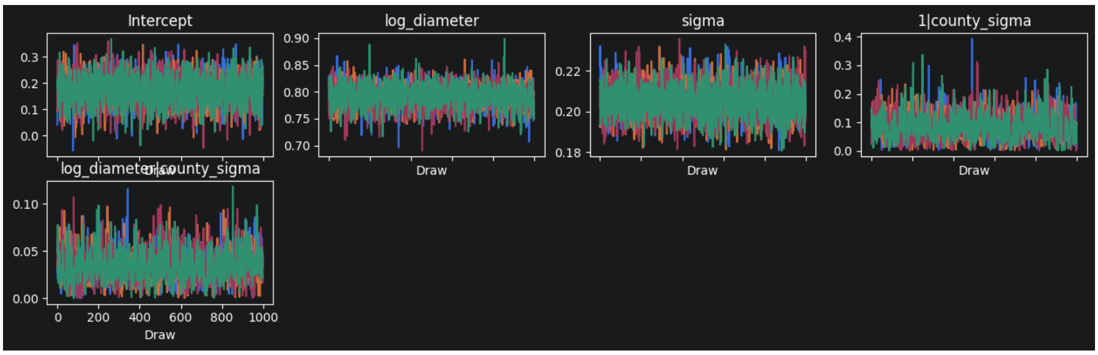

Figure 4: Traceplots for global effect coefficients, standard deviation and county-specific standard deviations in one-level hierarchical model with varying slopes.

Due to the large number of counties and plots, we did not create trace plots for each group-level effect (intercepts and slopes). However, the trace plots prepared for global coefficients and group-specific variances indicate that the corresponding chains have converged properly to the stationary distribution.

In order to check the convergence for all parameters, the potential scale reduction statistic $\hat{R}$ was calculated and compared with the value 1.05 as a rule of thumb. On this basis, we concluded that there were no convergence issues.

```
tabel = az.summary(results_cust_prior)
tabel[tabel['r_hat'] > 1.05]
```

mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat

Figure 5: An empty table of parameters for which $\hat{R} > 1.05$

To test the assumption of independence of the simulated sample, we prepared autocorrelation plots and estimated effective sample size, which measures how much independent information is contained in the posterior draws. Once again, this was done for the same parameters for which the trace plots were prepared.

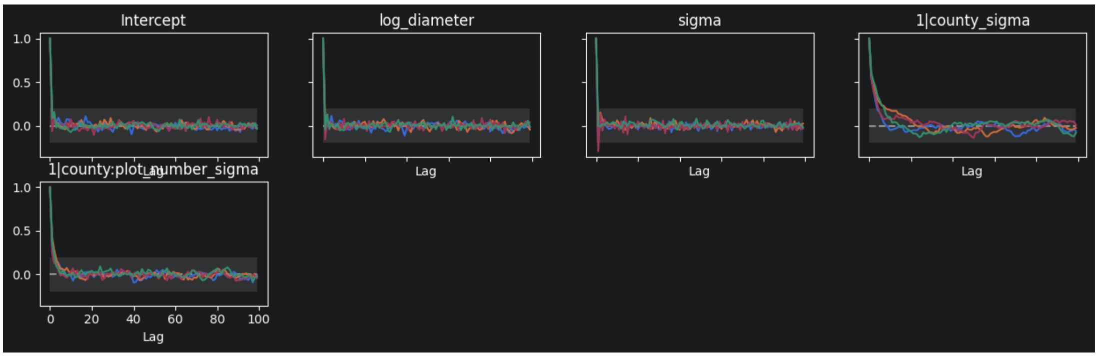

Figure 6: Autocorrelation plots for the two-level model with default priors. Empirical autocorrelations are close to 0.

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
Intercept	0.198	0.052	0.101	0.295	0.001	0.001	2650.0	2869.0	1.00
log_diameter	0.776	0.016	0.745	0.807	0.000	0.000	4561.0	3101.0	1.00
sigma	0.165	0.007	0.151	0.177	0.000	0.000	5139.0	3167.0	1.00
1|county_sigma	0.040	0.031	0.000	0.092	0.001	0.001	599.0	1357.0	1.01
1|county:plot_number_sigma	0.143	0.019	0.109	0.180	0.001	0.000	1156.0	2295.0	1.00

Figure 7: Effective sample size estimates for the two-level model with a custom prior for β.

As a rule of thumb it is recommended that the acceptable effective sample size should be greater than 100 times the number of chains [10]. In our case the number of chains is 4 and since every ESS estimate is greater than 400, we do not identify any problems regarding autocorrelation in our sample, despite considerably low ESS for the county-specific standard deviation, which, in turn, is consistent with the distinctive autocorrelation plot shown in Figure 6.

4.2 Posterior predictive checks

Now we will verify the models' fit to the data by analysing the simulated posterior predictive distribution. The approach presented below is based on the fact that the empirical posterior predictive distribution should have the same quantitative properties as the original distribution of the data.

For comparison, we use kernel density estimates of the original data (marked with a white line) and data simulated from the posterior predictive distribution (marked with blue lines).

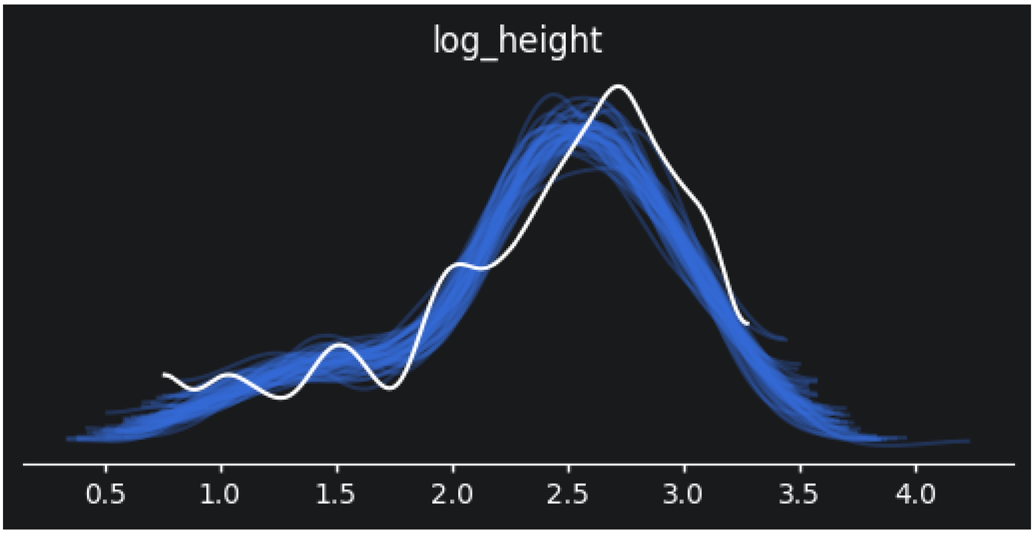

Figure 8: Kernel density estimates of the posterior predictive distribution and the original data for the simple non-hierarchical model.

As we can see, the overall structure of the original distribution has been preserved, though there are some substantial differences which may indicate that the model is not very adequate. Compared to the original distribution, the left tails of posterior predictive density estimates are flattened (this difference may be further amplified by the use of kernel estimation) and the mass along with the mode is shifted to the left. As a result, this could mean that the a posteriori estimates are underestimated and do not take into account variations in the number of shorter trees. However, at this point it should be noted that there are significantly less original observations from the $(0.7, 2.5)$ range on logarithmic scale, and that most of these values are rounded, which can be deduced by analysing scatter plots in Figure 2 (data points in this log-height range are arranged along straight lines). Contrary to the earlier conclusions, this might suggest that the model corrects errors related to flaws in our data.

To check the similarity between the empirical posterior predictive distribution and the original data distribution we also compute and compare values of auxiliary statistcs (such as mean, maximum and minimum) for posterior predictive samples and original data. The value for the original data should be typical among values for samples from posterior predicitve distribution.

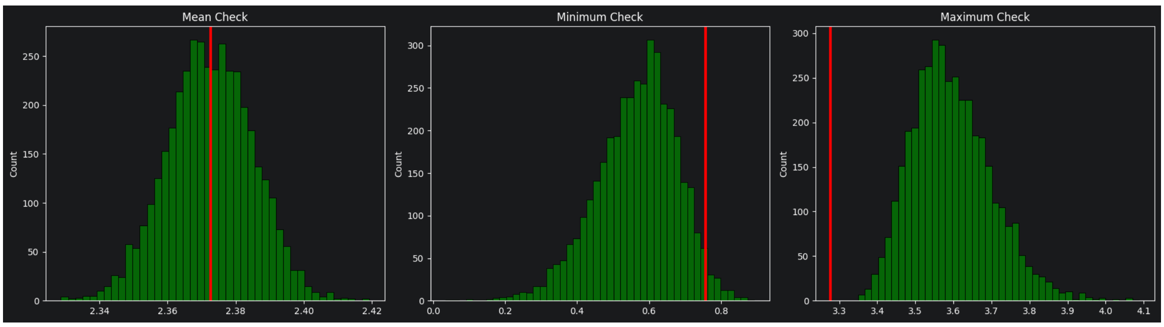

Figure 9: Test statistics for the original data and data simulated from the posterior predictive distribution for the two-level model with a cutom prior for β for the year 2022.

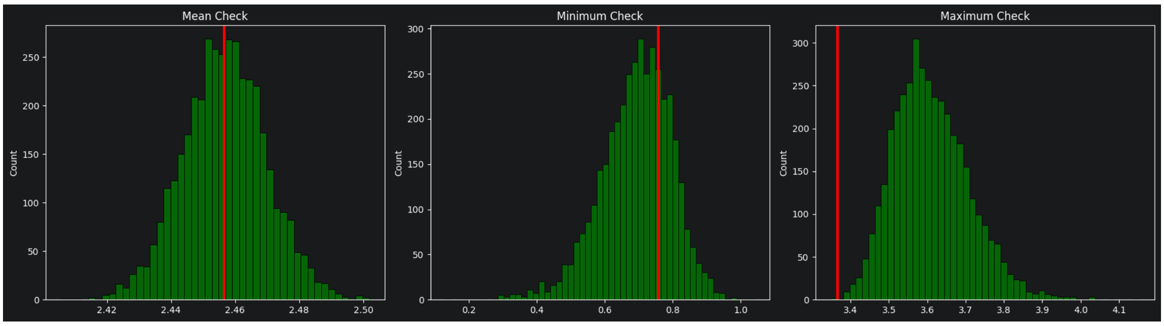

Figure 10: Test statistics for the original data and data simulated from the posterior predictive distribution for the two-level model with a cutom prior for β for the year 2017.

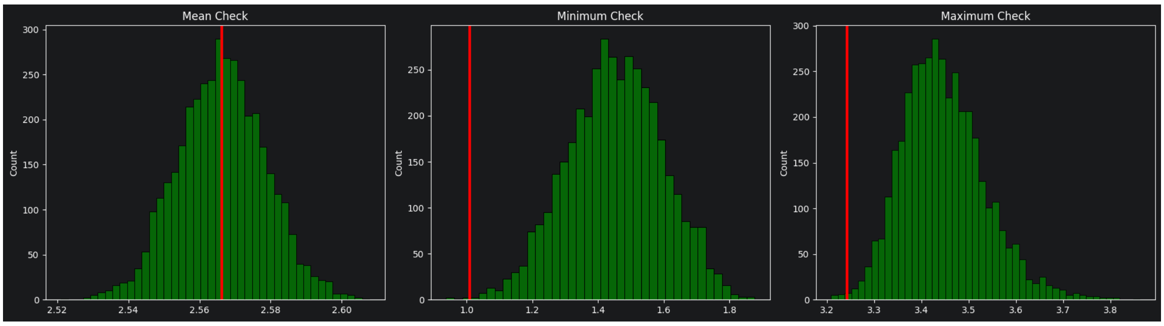

Figure 11: Test statistics for the original data and data simulated from the posterior predictive distribution for the two-level model with a cutom prior for β for the year 2012.

For each year, the results for the mean test statistic meet our expectations. In this aspect predicitve properties of the model are without fault. However, this is not the case for the maximum and minimum test statistics, which expose poor predictive power of the model for extreme values.

5 Results

5.1 Model comparison

As mentioned earlier, we decided to focus on the 'best' model among those presented in Section 3. To that end, we used the leave-one-out estimator of the expected log pointwise posterior predictive density, or ELPD_{loo} in abbreviated form.

```
The rhat statistic is larger than 1.01 for some parameters. This indicates problems during sampling. See https://arxiv.org/abs/1903.08008 for details
/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/arviz/stats/stats.py:782: UserWarning: Estimated shape parameter of Pareto distribution is
  warnings.warn(
/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/arviz/stats/stats.py:782: UserWarning: Estimated shape parameter of Pareto distribution is
  warnings.warn(
              rank    elpd_loo      p_loo  ...        dse  warning  scale
Custom_prior     0  106.863853  43.952006  ...   0.000000     True    log
Giant            1  106.570501  43.666417  ...   0.699555     True    log
County           2   51.214641  12.501628  ...  10.101821    False    log
Naive            3   28.721569   2.921625  ...  11.753089    False    log
```

Figure 12: Comparison of the models based on the ELPD_{loo} statistic.

Based on this criterion, we conclude that the two-level hierarchical model provides the best fit, with negligible differences regarding prior specification of this model. Warnings in the Figure 12 indicate that the MCMC convergence in the case of this model was not ideal, but we have already checked that $\hat{R} < 1.05$ for all parameters and assessed that the convergence is acceptable.

5.2 Posterior distribution

A particular advantage of the Bayesian approach is that it provides an entire probability distribution describing the possible values of the parameters as well as posterior predictive distribution, which allows us to make detailed predictions based on a given set of data. Additionally, the hierarchical form of the model takes into account differences between subgroups and thus gives us even more precise predictions.

However, the use of logarithmic transformation in our model requires us to exercise particular caution when interpreting our results. We will illustarte this by reffering to a generelised regression model in the following heuristic way. In this setting, analogous to the general formulation of our model, we are interested in modelling the conditional expectation of the target variable Y given the explanatory variable X.

$$Y = e^{\alpha} X^{\beta} e^{\varepsilon}, \quad \text{where} \quad e^{\varepsilon} \sim \mathrm{LogNormal}(0, \sigma).$$

Then

$$\mathbb{E}[Y|X] = e^{\alpha} X^{\beta} \cdot \mathbb{E}[e^{\varepsilon}] = e^{\alpha} X^{\beta} e^{\frac{\sigma^2}{2}} = e^{\alpha + \frac{\sigma^2}{2}} X^{\beta},$$

thus

$$\ln \mathbb{E}[Y|X] = \alpha + \frac{\sigma^2}{2} + \beta \cdot \ln X.$$

However

$$\mathbb{E}[\ln Y|X] = \alpha + \beta \cdot \ln X$$

and we conclude that if we want to obtain direct interpretation, a simple inverse transformation of the coefficient α will be an underestimation [11]. This could also be seen from Jensen's inequality for the concave function ln:

$$\mathbb{E}[\ln Y|X] \leqslant \ln \mathbb{E}[Y|X].$$

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
Intercept	0.198	0.052	0.101	0.295	0.001	0.001	2650.0	2869.0	1.00
log_diameter	0.776	0.016	0.745	0.807	0.000	0.000	4561.0	3101.0	1.00
sigma	0.165	0.007	0.151	0.177	0.000	0.000	5139.0	3167.0	1.00
1|county_sigma	0.040	0.031	0.000	0.092	0.001	0.001	599.0	1357.0	1.01
1|county:plot_number_sigma	0.143	0.019	0.109	0.180	0.001	0.000	1156.0	2295.0	1.00

Figure 13: Posterior summary for global coefficients and global and group-specific standard deviations for the two-level model with a custom prior for β.

So, eventually, after obtaining the estimates of the parameters $\hat{\alpha}$ and $\hat{\sigma}$, such as maximum a posteriori estimate (posterior mode) or the posterior mean, we can compute the corrected base value $e^{\hat{\alpha} + \frac{\hat{\sigma}^2}{2}}$, which corresponds to the height of a tree with diameter equal to 1cm. If we consider the posterior means given by the posterior summary shown in Figure 13, then this value is approximately equal to 1.24m, which is close to breast height and should not be concerning, since the height at which the 1cm diameter is measured is close to the height of the entire tree. Note that this problem along with its solution naturally extend to the group-specic intercepts in hierarchical models and the pointwise predicitons.

Also, observe that this problem does not affect β, which is the elasticity coeffient and has a natural interpretation, which was discussed in Section 3.1. Namely, if we base our estimate on the posterior mean, we obtain the following relationship: should the diameter increase by 10%, the height will increase by around 7.76%, (with the 94% credible interval being (7.46%,8.07%)), indicating that height gain is curbed as the tree grows thicker and thicker. It is therefore up to experts in biology and forestry to investigate and explain this phenomenon scientifically.

The hierarchical structure of the model allows us to compare the variation in environmental conditions at plot level with that at county level (since the place-specific intercepts in the model are intended to reflect initial place-specific conditions). Based on the posterior summary shown in Figure 13, the plot-specific standard deviation is greater than the county-specific standard deviation, which in itself could be regarded as insginificant, since values very close to 0 lie within the corresponding credible interval.

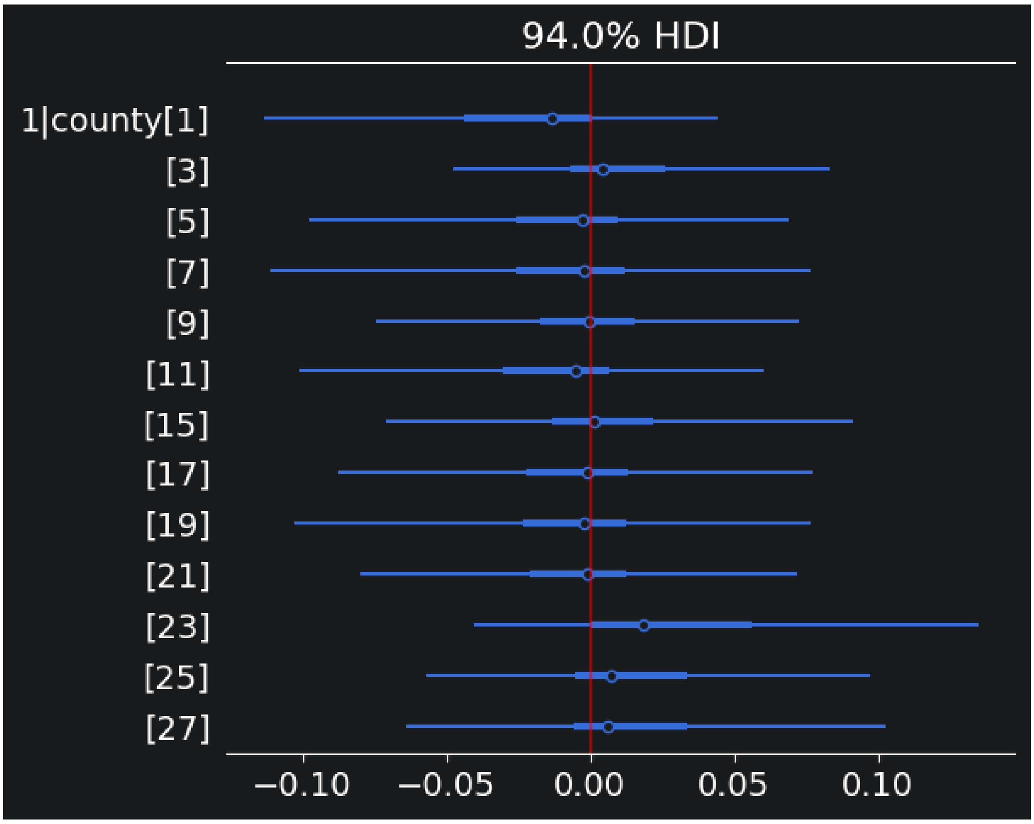

Figure 14: Posterior distribution of county-specific slopes summarized by credible intervals.

5.3 Predictions in the model

As established in the Problem Statement, we are also interested in the predictive properties of the model, which were partly examined in the Section 4.2. Similarly to what is shown in Figure 8, we can use density estimates to illustrate the posterior predicitve distribution for any given value of the regressor, that is, the diameter, or prepare a single plot with credible intervals summarizing posterior predictive distribution for diameter plotted on the x-axis. However, we should keep in mind that estimates of this distribution are based on the log-transformed model formulation and if we want to draw direct conclusions regarding the height, we ought to make adjustments analogous to the ones outlined in Section 5.2.

6 Summary

The main model considered in this project provides a satisfactory description of the relationship between height and diameter in the red spruce population in the state of Vermont. It enables us to estimate in great detail the percentage increase in height as a function of percentage increase in diameter, and to predict height based on diameter, taking into account the differences between individual parts of the study area.

7 Contribution and AI Statement

7.1 Contributions

- Gabriel Miziołek: conceptualization, report writing, presentation (mostly analytical and editorial aspects, such as structuring the document and interpreting the results, as well as the design and preparation of the final presentation materials)
- Fiodor Zianiucicz: conceptualization, data cleaning, model implementation (mostly technical aspects, such as programming, as well as the configuration and maintenance of the GitHub repository)

7.2 AI Statement

We used AI at the level of conceptualization and searching for the appropriate database (forest databases in Poland were hard-to-reach) as well as analysis of the model, with restricted reliance manifested in careful verification of the output.

The use of AI in the coding process was limited to preparing plots regarding auxiliary statistics for posterior predicitve checks and confidence intervals for plot-specific intercepts.

References

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